

SHANTANU

ECE Undergraduate | Embedded Systems Engineer (ESP32, Sensors, IoT)

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Linkedin | Portfolio | GitHub

EDUCATION

- **Thapar Institute of Engineering and Technology** Patiala, Punjab
Bachelor of Technology - Electronics and Communication Engineering August 2025 - Present

SKILLS SUMMARY

- **Embedded Systems:** ESP32, Arduino, I2C, UART, SPI, MQTT, Real-time Processing
- **Programming:** C, C++ (Arduino), Python, JavaScript
- **Hardware / Electronics:** Sensors, Electromagnets, Power Electronics, GPS Modules, OLED Displays
- **Tools / Platforms:** Git, GitHub, KiCad, Arduino IDE, ESP-IDF, Vercel

PROJECTS / OPEN-SOURCE

- **Zephyr-Station** ESP32, C++
Designed an ESP32-based environmental monitoring system integrating multiple sensors via I2C with OLED visualization, SD-card data logging, and threshold-based alert mechanisms.
- **Jolt-Locator** ESP32, C++
Developed a GPS-based true-north navigation system on ESP32 with real-time heading computation and location processing using UART communication.
- **The-Ruin-Machine** ESP32, C++
Implemented a probability-based stochastic simulator on ESP32 with physical I/O interfacing (OLED display and buzzer actuation).
- **Geometrical Shape Detection | Report** Python, OpenCV
Developed 6 image analysis algorithms using OpenCV and NumPy for automated geometrical shape detection and recognition with modular Python implementations.

EXPERIENCE

- **Ragastra MV Lev** Patiala, Punjab
Hardware & Controls Engineer September 2025 - Present
 - **Electromagnetic System Design:** Designed and tuned custom electromagnets for magnetic levitation prototype with optimized coil parameters.
 - **Power Electronics:** Designed and debugged SMPS-based power circuitry for inductive electromagnetic loads.
 - **Motor Control Systems:** Implemented Arduino-based synchronized dual stepper motor control with real-time actuation.
- **Grosity** Patiala, Punjab
Early-Stage Founding Engineer October 2025 - February 2026
 - **Web Development:** Developed and deployed company website using AI-assisted development workflows (Claude, GitHub Copilot, Kiro IDE).
 - **Autonomous Systems R&D:** Contributed to embedded electronics and control logic for autonomous payload-delivery drone systems.
 - **Product Strategy:** Aligned engineering decisions with B2B and D2C product models and business strategy.

ONGOING RESEARCH

- **GHFCVs - Reverse Engineering Study** December 2025 - Present
Literature review and comparative analysis of DC-DC converter topologies with hands-on teardown of commercial GHFCV systems to understand power electronics architecture and operational principles.

CERTIFICATIONS & ACHIEVEMENTS

- **Internet of Things** - MIT Professional Education x Santander Open Academy
- **Google: Artificial Intelligence and Productivity** - Google x Santander Open Academy
- **National Finalist** - E Summit Ideastorm 2026, IIT Roorkee